

Let's break down how GitHub Classroom works, focusing on the intuition behind each step:

### **The Core Idea:**

GitHub Classroom aims to streamline the distribution, collection, and grading of assignments using the power of Git and GitHub. It's designed to make the workflow smoother for both instructors and students.

### **Intuition Breakdown:**

#### **1. Template Repositories (The Assignment Blueprint):**

- Imagine you're a teacher handing out worksheets. Instead of physical paper, you create a "template repository" on GitHub.
- This repository contains all the starting files, instructions, and any necessary code for the assignment. It's like a pre-made folder with everything the students need to begin.
- **Intuition:** This ensures all students start with the same baseline, eliminating setup issues and providing a consistent foundation.

#### **2. Classrooms (The Virtual Classroom):**

- GitHub Classroom lets you create a "classroom" within your GitHub organization. This acts as a central hub for all assignments related to your course.
- **Intuition:** A classroom is a container for all the assignments and students, keeping everything organized within GitHub.

#### **3. Assignment Creation (Distributing the Worksheet):**

- When you create an assignment in GitHub Classroom, you link it to your template repository.
- GitHub Classroom then automatically creates a *private* repository for each student (or team) based on the template.
- **Intuition:** Each student gets their own personal copy of the assignment, just like handing out individual worksheets. This ensures students work independently and prevents accidental overwrites.

#### **4. Student Acceptance (Receiving the Worksheet):**

- Students receive a link to accept the assignment. When they click the link, GitHub Classroom creates their private repository and grants them access.
- **Intuition:** This is like the student saying, "Yes, I'll take this assignment." It initiates the process of creating their personal workspace.

#### **5. Student Work (Completing the Worksheet):**

- Students then clone their private repository to their local machines, complete the assignment, and push their changes back to GitHub.
- **Intuition:** This is the student doing the actual work on their own copy. Git acts as a version control system, tracking their progress and allowing them to save different versions of their work.

#### **6. Automatic Grading (Checking the Answers):**

- GitHub Classroom integrates with automated testing tools (like GitHub Actions) to provide automatic grading.
- You can set up tests that run whenever a student pushes their code.
- **Intuition:** This is like having a grading rubric that automatically checks if the student's work meets the requirements. It saves instructors time and provides instant feedback to students.

#### **7. Instructor Review (Providing Feedback):**

- Instructors can easily review student submissions directly on GitHub, providing feedback through comments, code reviews, and pull requests.
- **Intuition:** This allows instructors to give personalized feedback, just like writing comments on a physical worksheet. GitHub's features make it easy to point to specific lines of code and suggest improvements.

#### **8. Assignment Management (Keeping Track of Worksheets):**

- GitHub classroom, provides a dashboard that allows the instructor to see the status of all the assignments.
- **Intuition:** This is like a gradebook, that is always up to date. It gives the instructor a high level overview of the class progress.

**In essence, GitHub Classroom leverages the power of Git for:**

- **Distribution:** Creating individual copies of assignments.
- **Version Control:** Tracking student progress.
- **Collaboration (for team assignments):** Allowing students to work together.
- **Automation:** Automating grading and feedback.
- **Organization:** Keeping everything in one place.

Okay, let's break down how to manage your "Introduction to Vectorization in R" (Homework 1) assignment using GitHub Classroom. I'll provide a detailed, step-by-step guide.

### Prerequisites:

- **GitHub Account:** You already have this (rgt47).
- **GitHub Organization (Recommended):** It's best to organize your classroom work within a GitHub Organization. If you don't have one, create one now (e.g., "PHB228-Spring2025"). This helps keep your course repositories separate.
- **GitHub Classroom Account:** Go to [classroom.github.com](https://classroom.github.com) and sign in with your GitHub account. If this is your first time, you'll need to grant permissions.

### Step-by-Step Guide:

#### 1. Prepare the Assignment Template Repository:

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* **Create a Repository:**
  * On GitHub, create a **new repository**.
  * **Name it:** Something descriptive like `hw1-template` or `phb228-hw1-template`.
  * **Important:** **DO NOT** include any student-specific information in this repository. It's the
  * **Initialize with a README (Optional but Recommended):** A basic README is good practice. You can
  * **DO NOT add a .gitignore or License at this stage** (unless you have very specific needs for the
  * **Keep it PRIVATE:** While not strictly required, it's generally good practice to keep template
* **Add Assignment Files (If Necessary):**
  * In your case, the assignment asks students to create the `hw1_solutions.R` file. Since students
  * **Example Scenarios for Template Files:**
    * If you want to provide a skeleton R file with just the header comments, you could include an `
    * If there are any data files *other* than the built-in datasets (`iris`, `ChickWeight`) that s
    * If you have a helper function file, include that.
  * **For your specific assignment, the template repo could be very minimal, even just a README.**
* **Example Template Repo Structure (Minimal):**
  \\\
  hw1-template/
  |— README.md (Optional: Brief overview)
  \\\
* **Example Template Repo Structure (With Skeleton):**
  \\\
  hw1-template/
  |— hw1_solutions.R (Skeleton with header comments)
  |— README.md (Optional)
  \\\
* **Example `hw1_solutions.R` (Skeleton):**
  ```r
  # Your Name: [Your Name]
  # Course: PHB 228, Assignment 1
```

```

# Date: [Date]
# Description: This script provides solutions for Homework 1, focusing on R vectorization.

# Load required packages
library(tidyverse)
library(bench)

# Set seed for reproducibility
set.seed(123)

# ... (Students will add their code here)
` ``

```

## 2. Create the Assignment in GitHub Classroom:

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* **Go to GitHub Classroom:** classroom.github.com
* **Select Your Classroom:** Choose the organization you created (e.g., PHB228-Spring2025).
* **Create a New Assignment:** Click "New assignment."
* **Assignment Details:**
  * **Assignment Title:** "Homework 1: Introduction to Vectorization in R" (or similar)
  * **Deadline (Optional but Recommended):** Set the due date: "2025-04-09" and time: "11:59 PM". GitHub Classroom will remind students.
  * **Visibility:** Keep it private.
  * **Assign to:** Choose "Individual assignment" (since it's individual work).
  * **Select the Template Repository:** Find and select the `hw1-template` repository you created.
  * **Student Identifier:** How do you want to identify students? GitHub usernames are common and s
  * **Get started with a template repository:** Yes (you've already created it).
* **Click "Create assignment."**

```

## 3. Distribute the Assignment Link:

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* GitHub Classroom will generate an **assignment invitation URL**.
* **Share this link with your students** (e.g., through Canvas, your course website, or email).
* **Important:** This link is how students create their private repositories.

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## 4. Student Workflow:

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* **Student Clicks the Link:** They'll be prompted to accept the assignment.
* **Student Selects Themselves:** GitHub Classroom creates a private repository for that student, n
hw1-template` (using the GitHub username).
* **Student Clones the Repository:**
  * Students use Git to clone their repository to their local machines.
  * ``bash
    git clone https://github.com/PHB228-Spring2025/rgt47-hw1-template # Example URL
    ` ``
* **Student Completes the Assignment:**
  * Students write their R code in `hw1_solutions.R` (or any other required files).
* **Student Commits and Pushes Changes:**
  * ``bash
    git add hw1_solutions.R # Add the file
    git commit -m "Completed Homework 1" # Commit changes
    git push origin main # Push to GitHub
    ` ``

```

## 5. Instructor Review and Grading:

```

* **Go to GitHub Classroom:** classroom.github.com

```

\* \*\*Select Your Classroom and Assignment:\*\* You'll see a list of students and their repositories.

\* \*\*Review Student Submissions:\*\*

- \* Click on a student's repository to view their code.
- \* GitHub allows you to:
  - \* View the code directly in the browser.
  - \* Clone the repository yourself.
  - \* Leave comments on specific lines of code.

\* \*\*Provide Feedback:\*\* Use GitHub's commenting features or download the code to run it.

\* \*\*Record Grades:\*\* GitHub Classroom doesn't directly manage grades. You'll likely record them in y

### Important Considerations:

- **Communication:** Clearly communicate the workflow to your students. Provide a short tutorial or guide on using Git and GitHub (cloning, committing, pushing).
- **Git Basics:** Students *need* to understand basic Git commands.
- **Starter Code vs. Empty Template:** Decide if providing a skeleton `hw1_solutions.R` is helpful or if you want them to start completely from scratch.
- **Automated Testing (Optional but Powerful):** For more advanced workflows, explore GitHub Actions to automate grading of specific parts of the assignment (e.g., running tests on their functions). This requires more setup but can save time.
- **Late Submissions:** GitHub records the time of the last push, which can help you verify submission times (though you'll need to interpret it against your deadline).
- **Academic Integrity:** Reinforce your academic integrity policy. GitHub tracks commit history, which can sometimes reveal suspicious activity (e.g., identical commit histories).

Let me know if you'd like more detail on any of these steps, especially regarding setting up the template repository or automated testing!